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ese experiments were cared for able practices as outlined in the Use of Agricultural Animals in Agricultural Research and Teaching (FASS, 1999).

This experiment was designed to evaluate the effects of calcium salts of PUFA on serum concentrations of  $P_4$  and insulin in non-lactating, pregnant cows. This animal category was selected to avoid the partitioning of supplemented FA into milk, which could decrease the PUFA availability for body metabolism. Furthermore, the laboratory utilized ovariectomized cows inserted with an intravaginal device containing  $P_4$  as the model for evaluating  $P_4$  metabolism (Moriel et al., 2008; Lopes et al., 2009). However, because the experiment was conducted at a commercial dairy farm, ovariectomized cows were not an option. Thus, pregnant cows at later stages of pregnancy were selected in order to have a constant supply of  $P_4$  (e.g. corpus luteum- and placenta-derived  $P_4$ ), and hence, have a clearer picture of potential effects of PUFA supplementation on  $P_4$  metabolism. Forty-five Holstein animals (18 heifers and 27 multiparous cows; BW = 561±114 kg; BCS = 2.9±0.3; days pregnant = 110±56 d) were stratified by initial BW and BCS (Wildman et al., 1982). Cows were randomly assigned to receive individual daily (as-fed basis) supplementation of 0.50 kg of finely ground corn plus 0.22 kg of kaolin (a rumen-inert indigestible substance; CON), calcium salts of saturated fatty acids (SFA; Megalac®; Arm & Hammer Animal Nutrition, Division of Church & Dwight Co.; Inc.), or calcium salts of polyunsaturated fatty acids (PF; Megalac-E®; Quimica Geral do Nordeste, Rio de Janeiro, Brazil) (Table 1) for 14 d. The SFA treatment was included as an iso-lipidic, iso-caloric, and iso-nitrogenous control treatment relative to PF. Also, a supplementation period of 14 d was utilized based on previous findings (Lopes et al., 2009, 2011) observing positive effects of PUFA supplementation on reproductive performance of beef cows. Treatments were top-dressed daily (07.00 hours) and individually. Cows were offered free choice access to a corn silage-based diet (Table 2), in a dry lot facility with free access to Tifton (*Cynodon* spp) pastures for 14 d. Cows had free access to water throughout the experiment.

Blood samples were collected on days 0, 7 and 14, immediately prior to (0 h), and 3, 6, 9 and 12 h after the morning supplement feeding, for determination of serum

Serum concentrations of  $P_4$  and insulin were determined using a Coat-A-Count Kit (DPC Diagnostic Products Inc.; Los Angeles, CA) solid phase  $^{125}$ I RIA, previously validated for cattle (Moriel et al., 2008). The intra- and inter-assay CV were 1.8 and 0.1% for  $P_4$ , and 10.3 and 7.4% for insulin, respectively.

Blood measurements were analyzed as repeated measures using the MIXED procedure of SAS (Statistical Analysis System, version 9.2). The model statement contained the effects of treatment, time, day, parity and the resultant interactions. Cow (treatment × parity) was included as a random variable and error term for the tests of fixed effects. Cow (treatment × parity × day) was included as the subject in the repeated statement. The criterion for

Table 1 - Fatty acid (FA) composition (% of total FA<sup>1</sup>) of calcium salts of FA included in the SFA and PF supplements

|                         | Treatment       |                  |
|-------------------------|-----------------|------------------|
|                         | PF <sup>2</sup> | SFA <sup>3</sup> |
|                         | % of total FA   |                  |
| Lauric acid (12:0)      | 0.1             | 0.2              |
| Myristic acid (14:0)    | 0.2             | 1.6              |
| Palmitic acid (16:0)    | 17.5            | 50.8             |
| Palmitoleic acid (16:1) | 0.3             | 0.0              |
| Stearic acid (18:0)     | 5.1             | 4.1              |
| Oleic acid (18:1)       | 31.7            | 35.7             |
| Linoleic acid (18:2)    | 39.8            | 7.0              |
| Linolenic acid (18:3)   | 2.7             | 0.2              |
| Other                   | 2.6             | 0.4              |

PF - calcium salts of polyunsaturated fatty acids; SFA - calcium salts of saturated fatty acids.

<sup>1</sup> Values provided by manufacturers.

<sup>2</sup> Megalac-E (Quimica Geral do Nordeste, Rio de Janeiro, Brazil).

<sup>3</sup> Megalac (Arm & Hammer Animal Nutrition, Division of Church & Dwight Co.; Inc.).

Table 2 - Ingredients and nutritional composition of the base diet fed to cows

| Ingredients                                     | Basal diet (% of DM) |
|-------------------------------------------------|----------------------|
| Corn silage                                     | 56.3                 |
| Whole cottonseed                                | 6.90                 |
| Corn finely ground                              | 14.2                 |
| Soybean hulls                                   | 4.89                 |
| Soybean meal                                    | 9.79                 |
| Mineral/vitamin mix <sup>1</sup>                | 3.00                 |
| Panicum maxim cv. Mombasa                       | 4.88                 |
| Total                                           | 100.0                |
| Nutrient profile <sup>2</sup> , % of dry matter |                      |
| Dry matter                                      | 49.8                 |
| Total digestible nutrients                      | 68.7                 |
| Crude protein                                   | 17.0                 |
| Calcium                                         | 0.65                 |
| Phosphorus                                      | 0.42                 |

<sup>1</sup> Composition: 55% finely ground corn and 45% minerals (9.34% Ca; 1.56% P;

of freedom for all the tests of e separated using PDIFF in all was set at  $P \leq 0.05$ , and tendencies were determined if  $P > 0.05$  and  $\leq 0.10$ .

For comparison of  $P_4$  concentrations in cows with insulin above or below the median before feeding, data were analyzed with the UNIVARIATE procedure of SAS for median determination and the MIXED procedure of SAS to determine insulin effects on  $P_4$  concentrations. The model statement for this analysis contained the effects of insulin (above or below median), time, and the resultant interaction. For the analysis of insulin concentrations within each group, the model statement included only the effects of time. Period was the random variable and error term for the tests of fixed effect for both analyses within each group. Cow (period) was used as the subject in the repeated statement. Following the same criterion for choosing the covariance structure as mentioned above, the compound symmetry covariance structure was used in all analyses. Satterthwaite approximation was used to determine the denominator degrees of freedom for all the tests of fixed effects. Significance was set at  $P \leq 0.05$ , tendencies were determined if  $P > 0.05$  and  $P \leq 0.10$ . Results are reported according to main effects if no interactions were significant, or according to the highest-order interaction detected.

## Results

Dietary treatments did not affect ( $P \geq 0.52$ ) the overall serum concentrations of  $P_4$  and insulin (Table 3). Time effects were detected ( $P < 0.01$ ; Figure 1) for serum concentrations of  $P_4$  and insulin. Serum  $P_4$  concentrations decreased from 0 (6.09 ng/mL; SEM = 0.25) to 3 h (5.73 ng/mL; SEM = 0.25), but remained unaltered from 3 to 12 h (5.60 ng/mL; SEM = 0.25) after feeding the supplements. Serum concentrations of insulin peaked at 3 h (12.51  $\mu$ IU/mL; SEM = 0.93), but decreased 12 h after supplements were offered (7.69  $\mu$ IU/mL; SEM = 0.98).

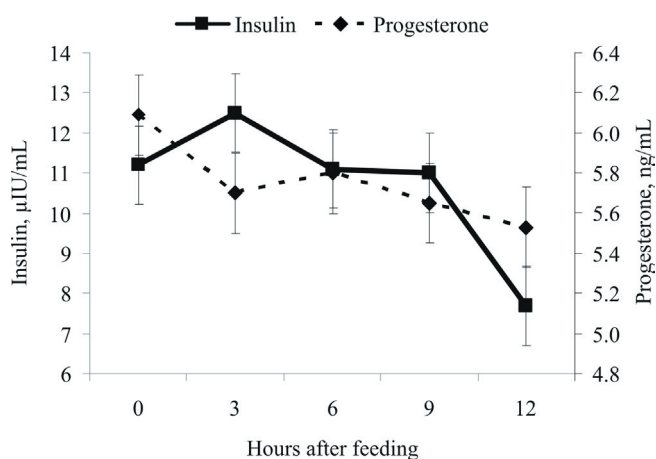
A parity effect was detected ( $P = 0.04$ ; Table 4) for serum  $P_4$  concentrations. Heifers had greater serum concentrations of  $P_4$  compared with multiparous cows (6.35 vs. 5.16 ng/mL, respectively; SEM = 0.42). Similarly, a parity effect ( $P < 0.001$ ; Table 4) was detected on serum concentrations of insulin because multiparous cows had greater ( $P < 0.01$ ) insulin

median (10.41  $\mu$ IU/mL) immediately before the morning feeding. Cows having insulin concentrations above the median had similar ( $P = 0.15$ ) mean concentrations of  $P_4$  compared with cows having insulin concentrations below the median (5.64 vs. 5.87 ng/mL, respectively; SEM = 0.25; Figure 2). There was a tendency for treatment  $\times$  time effect ( $P = 0.06$ ) on serum concentrations of  $P_4$ . Cows having

Table 3 - Overall serum concentrations of progesterone and insulin of cows fed rumen-inert PUFA or SFA

| Item                 | Treatment |      |      | SEM  | P-value |
|----------------------|-----------|------|------|------|---------|
|                      | CON       | SFA  | PF   |      |         |
| Progesterone, ng/mL  | 5.52      | 6.13 | 5.63 | 0.41 | 0.52    |
| Insulin, $\mu$ IU/mL | 11.5      | 10.5 | 10.1 | 1.43 | 0.76    |

CON - Cows fed daily 0.50 kg of finely ground corn plus 0.22 kg of a rumen-inert indigestible substance (kaolin); SFA - Cows fed daily 0.50 kg of finely ground corn plus 0.22 kg of calcium salts of short-chain fatty acids; PF - Cows fed daily 0.50 kg of finely ground corn plus 0.22 kg of calcium salts of polyunsaturated fatty acids. SEM - standard error of the mean.



Within each sampling time (0, 3, 6, 9 and 12 h), serum concentrations of  $P_4$  and insulin were grouped by treatment and day of collection (d 0, 7 and 14). Cows were randomly assigned to receive individual supplementation of 0.50 kg of finely ground corn plus 0.22 kg kaolin (CON), calcium salts of saturated FA (SFA), or calcium salts of PUFA (PF) for 14 d.

Figure 1 - Serum concentrations of progesterone ( $P_4$ ) and insulin of non-lactating, pregnant dairy cows.

Table 4 - Serum concentrations of insulin and progesterone of heifers and multiparous cows

| Item                 | Parity <sup>1</sup> |      | SEM <sup>2</sup> | P-value |
|----------------------|---------------------|------|------------------|---------|
|                      | Heifers             | Cows |                  |         |
| Progesterone, ng/mL  | 6.35                | 5.16 | 0.42             | 0.04    |
| Insulin, $\mu$ IU/mL | 7.0                 | 14.4 | 1.49             | < 0.01  |

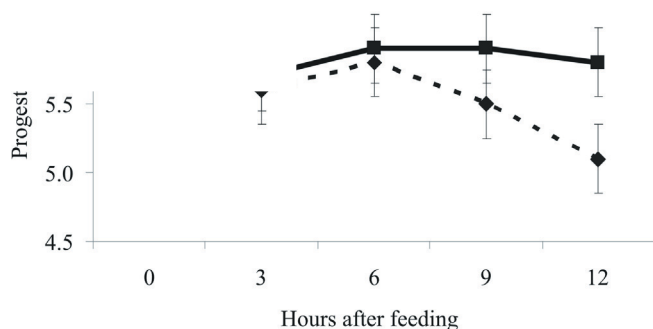


Figure 2 - Serum concentrations of progesterone ( $P_4$ ) of non-lactating, pregnant dairy cows having serum concentrations of insulin above or below the median (10.41  $\mu$ IU/mL) immediately before the morning feeding.

insulin concentrations above the median had lower ( $P < 0.01$ ) concentrations of  $P_4$  12 h after the supplements were offered compared with cows having insulin concentrations below the median (5.14 vs. 5.84 ng/mL, respectively; SEM = 0.25).

## Discussion

Polyunsaturated fatty acids play an important role on stimulation and inhibition of prostaglandin production (Staples et al., 1998), affecting ovary function and follicular development (Thatcher et al., 2004), improving the oocyte quality (Fouladi-Nashta et al., 2007), and increasing the first-service conception rate (Thatcher et al., 2006; Lopes et al., 2009). Although there is evidence that feeding fatty acids improved the energy status of lactating dairy cows (Staples et al., 1998), an enhanced reproductive performance may occur apart from an improved energy status of the animals (Sklan et al., 1991). In agreement, Reis et al. (2012) reported that feeding calcium salts of polyunsaturated fatty acids (PUFA) at 1.1% of diet DM to lactating dairy cows did not affect DM or energy intake, but decreased pregnancy losses compared with control cows that did not receive PUFA supplementation.

Adding calcium soaps of FA to the diet increased serum concentrations of  $P_4$  in bovine females (Hightshoe et al., 1991; Hawkins et al., 1995). Greater concentrations of  $P_4$  before and after breeding are associated with enhanced fertility (Folman et al., 1973) and embryonic survival (Mann et al., 2006). However, the present study showed no effects of

3) frequency of blood sampling.

Approximately 3 to 4 h following feeding, there is an increase in the rate of blood flow to the liver. This event leads to an increase in the metabolism of  $P_4$  by hepatic cells (Sangsritavong et al., 2002), which results in decreased concentrations of  $P_4$  after feeding (Sangsritavong et al., 2002; Vasconcelos et al., 2003). This response was observed in the present study, since all treatments had a decrease in  $P_4$  concentrations at approximately 3 to 4 h following feeding (Figure 1). Hepatic clearance of  $P_4$  may be diminished by adding PUFA to the diet. Sangsritavong et al. (2002) reported increased hepatic  $P_4$ -half life, and decreased clearance of  $P_4$  in non-lactating dairy cows infused with an emulsion of soybean oil. Piccinato et al. (2010) reported decreased hepatic clearance of  $P_4$  *in vitro* when concentrations of LNA were 100 to 300  $\mu$ M. However, plasma concentrations of  $P_4$  did not increase after *in vivo* infusions of 490 mL of linseed oil (56% LNA and 16.4% LA) in lactating dairy cows. Those authors concluded that the plasma concentrations of LNA (less than 8  $\mu$ M) after the linseed oil infusions were not sufficient to inhibit the key enzymes responsible for  $P_4$  metabolism (P450 2C and P450 3A) and thus, did not decrease hepatic clearance of  $P_4$ . Hawkins et al. (1995) reported increased serum concentrations of  $P_4$  in pregnant heifers when calcium salts of FA were fed at 0.57 kg/heifer daily. Fouladi-Nashta et al. (2007) only observed a tendency for greater  $P_4$  concentrations when lactating dairy cows were fed 0.80 kg/d of calcium salts of FA compared with 0.20 kg/d. The results of the present study are in agreement with Lopes et al. (2009) and Reis et al. (2012), who did not observe increase in serum  $P_4$  concentrations by adding calcium salts of PUFA at approximately 0.20 kg/cow daily. Taking these results together, it is possible that adding 0.22 kg of PF did not result in sufficient amounts of PUFA to decrease the hepatic clearance of  $P_4$ .

Duration of the supplementation period is another potential reason for the discrepancy between the results of the present study and others (Hightshoe et al., 1991; Hawkins et al., 1995). In the present study, a supplementation period of 14 d was used based on previous studies from our laboratory (Lopes et al., 2009, 2011). Hawkins et al. (1995) fed calcium salts of FA to pregnant heifers starting 100 d before parturition through day 12 of the third estrus cycle. Hightshoe et al. (1991) fed calcium salts of FA to

e fed 0.10 kg/d of rumen-inert ed with feeding a non-fat control on  $P_4$  concentrations when same rumen-inert PUFA were provided at 0.20 kg/d. The reason for 0.10 kg/d of rumen-inert PUFA resulting in greater concentrations of  $P_4$  in the study of Lopes et al. (2009) remains unclear. The less frequent blood sampling in the present study could be one of the reasons for the lack of differences detected in the circulating serum concentrations of  $P_4$ . Hawkins et al. (1995) collected blood samples at -60, -30 and 0 min before ovariectomy and 15, 30, 45, 60, 90 and 120 min after ovariectomy. Thereafter, blood samples were collected hourly for 5 h, and every other hour for an additional 6 h. Lopes et al. (2009) collected blood samples immediately before feeding (0 h), and 2, 4, 6, 8, 10 and 12 h after feeding. A more frequent blood sampling could have increased the chances of detecting treatment effects on circulating concentrations of  $P_4$ .

Although Piccinato et al. (2010) demonstrated that different concentrations of palmitic (16:0), palmitoleic (16:1), oleic (18:1) and LNA (18:3) inhibited the *in vitro* metabolism of  $P_4$ , the concentrations of each FA required to inhibit the hepatic clearance of  $P_4$  represents supraphysiological concentrations if utilized in *in vivo* studies. Therefore, the relative contribution of FA (other than LNA and LA) can be considered inexistent due to the small amount of calcium salts of FA provided in this study (0.22 kg/cow daily).

The effects of fat supplementation on insulin concentrations are varied, showing increase (Blum et al., 1985; Reis et al., 2012), no effect (Fouladi-Nashta et al., 2007; Reis et al., 2012), and decrease in concentrations of insulin after fat supplementation (Relling and Reynolds, 2007). Insulin concentrations reflect energy intake, and therefore, differences in insulin concentrations can be due to differences in energy status (Lucy et al., 1991). In the present study, no effects of rumen-inert PUFA supplementation were detected on serum concentrations of insulin, which suggests that differences in DMI did not occur. Likewise, Lopes et al. (2009) did not observe differences in serum concentrations of insulin in beef cows supplemented with up to 0.20 kg/d rumen-inert PUFA. Reis et al. (2012) reported that adding calcium salts of PUFA (1.1% of diet DM) increased serum  $P_4$  concentrations, and tended to

evaluations are required to confirm this hypothesis.

Circulating concentrations of  $P_4$  are highly affected by feed intake (Vasconcelos et al., 2003; Cooke et al., 2007; Moriel et al., 2008). The concentration of insulin peaked 3 h after feeding, whereas those of  $P_4$  decreased after feeding (Figure 1). The time effects detected for  $P_4$  concentrations were similar to previous reports (Vasconcelos et al., 2003; Cooke et al., 2007; Moriel et al., 2008). In the present study, pregnant heifers had greater serum concentrations of  $P_4$  compared with pregnant mature cows. In agreement with these results, heifers had greater serum concentrations of  $P_4$  from days 5 to 12 of the estrus cycle compared with lactating cows, despite their lower luteal volume on day 14 of the estrus cycle (Wiltbank et al., 2001). Sangsritavong et al. (2002) reported similar absolute increase in liver blood flow (LBF) of lactating and non-lactating dairy cows. However, as a percentage of basal LBF, non-lactating cows had greater increase in LBF compared with lactating cows, probably due to the lower basal LBF. Although the DMI of non-lactating pregnant heifers and cows estimated by the NRC (2001) are similar (as a percentage of BW; 2.0% of BW), the absolute amount of feed intake is lower for heifers than non-lactating cows. Thus, it is possible that LBF in non-lactating multiparous cows was greater (as a percentage of basal LBF), which consequently may have increased hepatic clearance of  $P_4$  and reduced  $P_4$  concentrations compared with heifers. However, further evaluation is required to confirm this hypothesis. In addition, although DMI was not measured in this study, greater DMI of non-lactating cows would also suggest greater energy intake and likely greater insulin concentration.

Previous results from our laboratory have shown that peripheral concentrations of insulin potentially modulates the hepatic metabolism of  $P_4$  (Moriel et al., 2008; Lopes et al., 2009; Vieira et al., 2010, 2013). Elevated insulin concentrations reduced hepatic expression of  $P_4$  catabolic enzymes P450 2C and P450 3A (Lemley et al., 2008; Vieira et al., 2013), which may further increase circulating concentrations of  $P_4$  (Moriel et al., 2008; Lopes et al., 2009; Vieira et al., 2010). Accordingly, Vieira et al. (2013) reported that dairy cows in adequate nutritional status receiving jugular glucose infusion (3 h at a rate of 0.5 g/kg of BW) had greater circulating insulin concentrations and reduced hepatic mRNA expression of  $P_4$  catabolic enzymes



g calcium salts of PUFA (1.1% m P<sub>4</sub> concentrations, and tended concentrations in primiparous, but not multiparous cows. In the present study, there was a tendency (Figure 2) for lower serum concentrations of P<sub>4</sub> in cows having serum insulin concentrations greater than the median. Moriel et al. (2008) also reported that cows with insulin concentrations above the median experienced a more severe decrease in P<sub>4</sub> concentrations 5 h after feeding. Those authors attributed this response to increased rumen fermentation, propionate production, and subsequent glucose synthesis, which likely caused the initial elevated concentrations of insulin concentrations but also enhanced hepatic blood flow after feeding. Thus, the increased metabolism of P<sub>4</sub> likely overrides a potential inhibitory effect of insulin on P<sub>4</sub> catabolism (Moriel et al., 2008).

## Conclusions

Supplementation of relatively small amounts of calcium salts of PUFA (0.22 kg) is not sufficient to enhance the serum P<sub>4</sub> or insulin concentrations of non-lactating, pregnant dairy cows. However, increasing the serum insulin concentrations seems to augment serum concentrations of P<sub>4</sub>. Therefore, better understanding of how nutritional management can affect insulin synthesis and the rate of hepatic P<sub>4</sub> metabolism may improve reproductive efficiency in dairy herds, but further evaluations are required.

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